DESIGN REFERENCE DOCUMENT · PART A

Web Flow Guide A to Z

TransitOps — Smart Transport Operations Platform. Complete site map, every screen, every button and exactly what it does, role-based navigation, and the full lifecycle logic that drives the app end-to-end.

PROJECT	DOCUMENT TYPE	BASED ON	VERSION
TransitOps	**Web Flow (A–Z)**	**PRD v1.0 & TRD v1.01**	**1.1**

PART A — WEB FLOW

Information Architecture & Site Map

TransitOps is a role-aware, multi-page web application. Every screen maps directly to a functional requirement in the PRD and a route group in the TRD. Navigation is persistent (top bar / sidebar); visibility of each link is governed by the signed-in user's role. No screen is more than one click from the nav bar — there is no deep nesting.

Top-Level Flow

Login / Auth → **Dashboard (role-aware KPIs)**

→ **Vehicle Registry** | **Driver Management** | **Trip Management** | **Maintenance** | **Fuel & Expense** | **Reports & Analytics**

Screen Inventory

Screen	Route	Purpose	Primary Role
Login	index.html	Email/password authentication — entry point for all users	All
Dashboard	dashboard.html	KPI cards, fleet utilization, filters by type/status/region	All
Vehicle Registry	vehicles.html	Register, edit, retire vehicles; view live status	Fleet Manager
Driver Management	drivers.html	Maintain driver profiles, license validity, safety scores	Safety Officer
Trip Management	trips.html	Create, dispatch, complete, cancel trips	Dispatcher
Maintenance	maintenance.html	Raise & close maintenance records per vehicle	Fleet Manager
Fuel & Expense	fuel-expense.html	Log fuel purchases and other operational expenses	Financial Analyst
Reports & Analytics	reports.html	Fuel efficiency, utilization, operational cost, ROI, CSV export	Financial Analyst

Role → Navigation Visibility Matrix

Nav Item	Fleet Manager	Dispatcher	Safety Officer	Financial Analyst
Dashboard	✔ view	✔ view	✔ view	✔ view
Vehicle Registry	Full CRUD	View only	—	View only
Driver Management	—	View only	Full CRUD	—
Trip Management	View only	Full CRUD + actions	View only	—
Maintenance	Full CRUD	—	—	View only
Fuel & Expense	—	—	—	Full CRUD
Reports	View only	—	—	Full access + export

PART A — SCREEN-BY-SCREEN (A → Z)

1. Login / Auth

Single entry point for every role. Nothing beyond this screen is reachable without a valid session.

Login ALL ROLES

`index.html`

Authenticates the user and routes them into a role-scoped Dashboard. This is the "A" of the A–Z journey — every session starts here.

Element	What it does	Where it goes / result
Email field	Captures the user's registered email address	Required before submit is enabled
Password field	Captures the user's password (masked)	Required before submit is enabled
Log In	Validates credentials against the server; determines role (Fleet Manager / Dispatcher / Safety Officer / Financial Analyst)	**Success →** redirects to `dashboard.html` with nav bar pre-filtered to that role's permissions. **Failure →** inline error message shown, user stays on Login

> Every subsequent screen in this document assumes the user has already logged in and the nav bar shows only the modules their role is entitled to (see Role → Navigation Visibility Matrix, page 2).

2. Dashboard

The landing screen after login. Every role sees the same layout — KPI cards + filters — but the numbers and which KPI cards appear are scoped to what that role is allowed to see.

Dashboard ALL ROLES

`dashboard.html`

Answers "what needs my attention right now" using role-aware KPI cards, fleet utilization, and filters by vehicle type, status, and region.

Element	What it does	Where it goes / result
KPI Cards (Active Vehicles, Fleet Utilization, Active Trips, Drivers on Duty)	Read-only live counters pulled from Vehicle Registry, Trip Management, and Driver Management modules	Clicking a card jumps to the related module (e.g. "Active Trips" → Trip Management, filtered to active)
Filter bar (type / status / region)	Narrows the KPI cards and any summary tables on this screen	Stays on Dashboard; re-renders cards with filtered data
Persistent nav bar	Lists only the modules the logged-in role can access	Each item routes to its respective screen (see matrix, page 2)

3. Vehicle Registry

Fleet Manager's home base for onboarding and managing every vehicle. Dispatcher and Financial Analyst can view but not edit.

Vehicle Registry

FLEET MANAGER — FULL CRUD

`vehicles.html`

Register, edit, and retire vehicles; view live status of the entire fleet.

Element	What it does	Where it goes / result
+ Register Vehicle	Opens the "Add Vehicle" form (Registration Number, Model, Type, Capacity)	On save → new row is added to the registry table with status set to ●Available
View Details (row action)	Opens the full vehicle detail panel	Shows edit form + status history + linked maintenance records
Edit (inside detail panel)	Updates vehicle attributes (model, capacity, type)	Saves changes; status is not editable here — it is system-driven
Retire action	Permanently removes the vehicle from active operations	Vehicle status → ●Retired — a terminal state, permanently excluded from every dispatch dropdown
Status pill (per row)	Read-only indicator — reflects live state, not editable directly	Available / On Trip / In Shop / Retired — driven by Trip & Maintenance modules

A vehicle's status is never changed by hand on this screen — it only changes as a side-effect of actions taken in Trip Management (Dispatch → On Trip) or Maintenance (Raise Record → In Shop).

4. Driver Management

Safety Officer's compliance console. Dispatcher can view driver info but cannot edit or suspend.

Driver Management SAFETY OFFICER — FULL CRUD

`drivers.html`

Maintains driver profiles, license validity, and safety scores; the compliance gate for who is allowed to drive.

Element	What it does	Where it goes / result
Driver table (license expiry, safety score columns)	Lets the Safety Officer scan for expiring licenses or low safety scores at a glance	Sorting/filtering stays on this screen
Suspend Driver	Flags a non-compliant driver (expired license or unsafe score)	Driver status → ●Suspended / Off Duty — instantly excluded from the Dispatcher's driver-selection list on Trip Management
Reactivate action	Clears a suspension once the driver becomes compliant again	Driver status → ●Available
Add / Edit driver profile	Creates or updates a driver record (license number, expiry date, contact info)	Saves to Driver Management table

5. Trip Management

The Dispatcher's operational cockpit — the busiest screen in the app, driving three separate lifecycles (Trip, Vehicle, Driver) at once.

Trip Management

DISPATCHER — FULL CRUD + ACTIONS

`trips.html`

Create, dispatch, complete, or cancel trips.

Element	What it does	Where it goes / result
+ New Trip	Opens the trip-creation form	Goes to vehicle + driver selection step
Select Vehicle + Driver dropdowns	Lists only vehicles with status Available and drivers with status Available (In Shop / Retired vehicles and Suspended / expired-license drivers are auto-filtered out)	Blocked with an inline error if either list is empty
Cargo weight / distance fields	Captures trip parameters	Validated against the selected vehicle's capacity
Save as Draft	Stores the trip without committing the vehicle/driver	Trip status → **Draft** (vehicle/driver remain Available)
Dispatch	Commits the trip to execution — **blocked** if cargo exceeds vehicle capacity or the driver's license is invalid/expired	Trip → **●Dispatched**; Vehicle → **●On Trip**; Driver → **●On Trip**
Cancel	Available at Draft or Dispatched stage, any time before completion	Trip → **Cancelled**; releases vehicle + driver back to **●Available**
Odometer + fuel fields	Captured at trip end	Required before Complete Trip can be pressed
Complete Trip	Closes out a Dispatched trip	Trip → **Completed**; Vehicle → Available; Driver → Available
Monitor on Dashboard	Navigational link, not an action	Jumps back to Dashboard to watch the trip's live status

A vehicle in "In Shop" status is automatically removed from every dispatch dropdown across the app until its maintenance record is closed. A suspended or expired-license driver is instantly excluded the same way.

6. Maintenance

Fleet Manager's workshop log — every record here directly controls whether a vehicle is dispatchable.

Maintenance FLEET MANAGER — FULL CRUD

`maintenance.html`

Raise and close maintenance records per vehicle.

Element	What it does	Where it goes / result
Raise Record	Opens a new maintenance record against a selected vehicle	Record → **Open**; Vehicle → **In Shop** (removed from every dispatch dropdown)
Close Record	Marks the maintenance work as finished	Record → Closed; Vehicle → **Available** — **unless** the vehicle was separately marked Retired, in which case it stays Retired

7–8. Fuel & Expense / Reports & Analytics

The Financial Analyst's two screens — logging cost data, then turning it into decisions.

Fuel & Expense

FINANCIAL ANALYST — FULL CRUD

`fuel-expense.html`

Log fuel purchases and other operational expenses.

Element	What it does	Where it goes / result
+ Log Entry	Opens the expense-entry form (vehicle, date, category, amount)	Saves a row to the expense ledger; feeds Reports & Analytics

Reports & Analytics

FINANCIAL ANALYST — FULL ACCESS + EXPORT

`reports.html`

Fuel efficiency, utilization, operational cost, and ROI, with CSV export. Other roles (e.g. Fleet Manager) get View only.

Element	What it does	Where it goes / result
Filter by vehicle / date	Narrows the report tables and charts	Stays on Reports; re-renders with filtered data
Export CSV	Packages the currently filtered view into a downloadable file	Triggers a browser file download — no navigation

PART A — WEB FLOW

Role-Based User Journeys (End-to-End)

Each persona follows a distinct primary path through the application, click by click. These are the backbone for wireframing and QA test-case design.

DISPATCHER — Create & Run a Trip

Login → Dashboard → Trips: + New Trip → Select available Vehicle + Driver → Enter cargo weight, distance → Save as Draft → Dispatch → Monitor on Dashboard → Enter odometer + fuel → Complete Trip (or Cancel at any point before completion)

System blocks the flow at "Select Vehicle + Driver" if either is unavailable, and at "Dispatch" if cargo exceeds capacity or the driver's license is invalid/expired.

FLEET MANAGER — Register a Vehicle & Handle Maintenance

Login → Dashboard → Vehicle Registry: + Register Vehicle → Status: Available → Maintenance: Raise Record → Vehicle → In Shop → Close Record → Vehicle → Available

A vehicle in "In Shop" status is automatically removed from every dispatch dropdown across the app until maintenance is closed.

SAFETY OFFICER — Monitor Driver Compliance

Login → Dashboard → Driver Management → Review license expiry / safety score → Suspend if non-compliant

A suspended or expired-license driver is instantly excluded from the Dispatcher's driver selection list.

FINANCIAL ANALYST — Track Cost & Export Reports

Login → Dashboard → Fuel & Expense: Log entry → Reports → Filter by vehicle/date → Export CSV

Core Lifecycle Diagrams

These state machines are enforced server-side and must be visually reflected in the UI (badges, disabled buttons, filtered dropdowns) at every screen listed above.

Trip Lifecycle

Draft → **Dispatched** → **Completed**

Draft → **Cancelled**

Dispatched → **Cancelled** (releases vehicle + driver back to Available)

Vehicle Status

Available dispatch → **On Trip** complete/cancel → **Available**

Available open maintenance → **In Shop** close → **Available***

Retired — terminal state, permanently excluded from dispatch

**Unless the vehicle was marked Retired — in that case it stays Retired even after maintenance closes.*

Driver Status

Available dispatch → **On Trip** complete/cancel → **Available**

Available admin action → **Suspended / Off Duty**

Maintenance Lifecycle

Record Created → **Open** (vehicle → In Shop) → **Closed** (vehicle → Available, unless Retired)